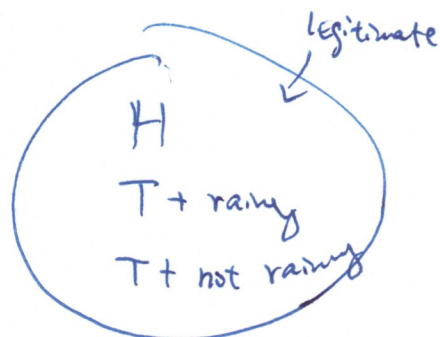


Lecture 1. Probability Models and Axioms.

1. Sample Space Ω

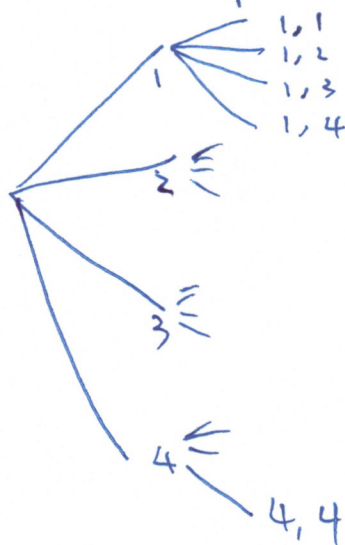
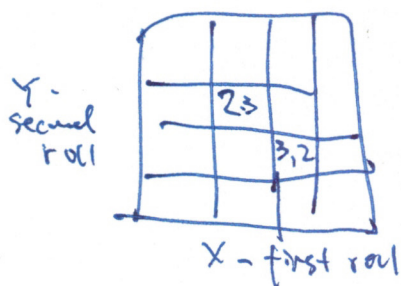
1. - set of possible outcomes.
2. - mutual exclusive
- collective exhaustive.



Sample space.

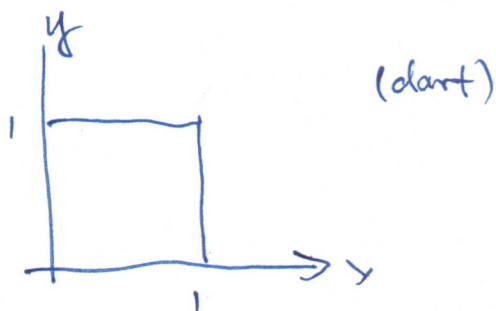
. two row of a tetrahedral die.

- sample space vs. sequential description



Sample space: Continuous example

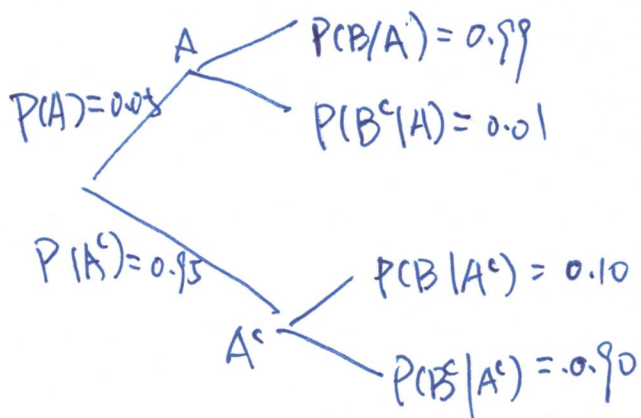
$$\Omega = \{(x, y) \mid 0 \leq x, y \leq 1\}$$



Ex. Model based on conditional probabilities

• Even A: Airplane is flying above.

Even B: Something register on radar screen.

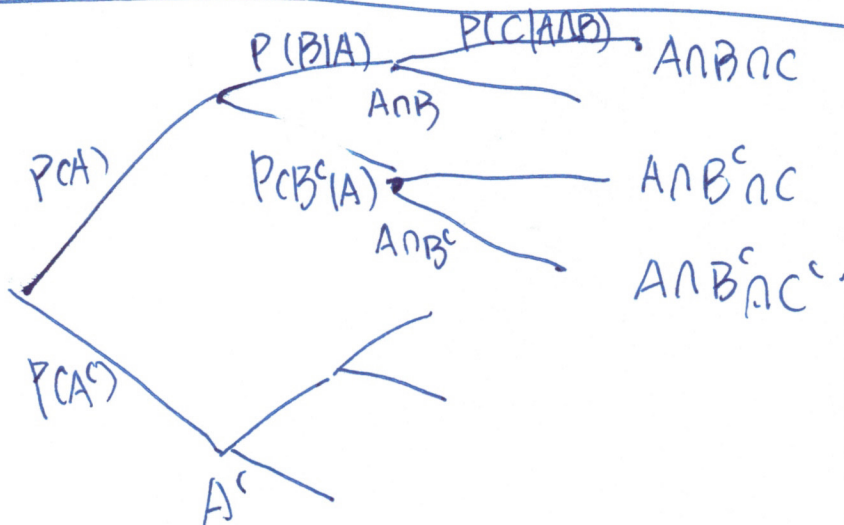


$$P(A \cap B) = P(B|A)P(A) = 0.05 \cdot 0.99 = 0.0495$$

$$P(B) = P(B|A) + P(B|A^c) = 0.049 + P(A^c) \cdot P(B|A^c) = 0.1445$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.0495}{0.1445} = 0.34$$

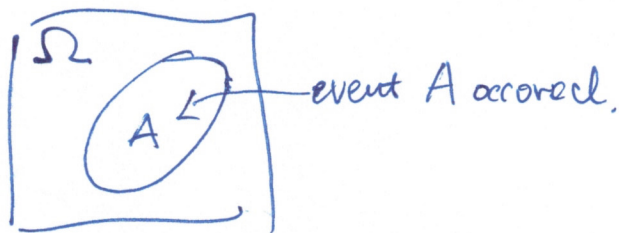
← counter intuitive:
a lot of times the B is false
alarm.



$$\begin{aligned} P(A \cap B \cap C) &= P(A \cap B)P(C|A \cap B) \\ &= P(B|A)P(A)P(C|A \cap B) \end{aligned}$$

probability axioms.

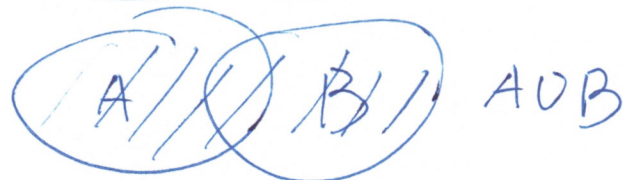
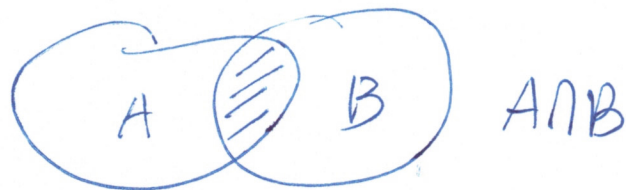
- **Event**: subspace in sample space
- probability is assigned to events
outcome is a point,



Axioms:

1. Nonnegativity: $P(A) \geq 0$
2. Normalization: $P(\Omega) = 1$
3. Additivity: If $A \cap B = \emptyset$

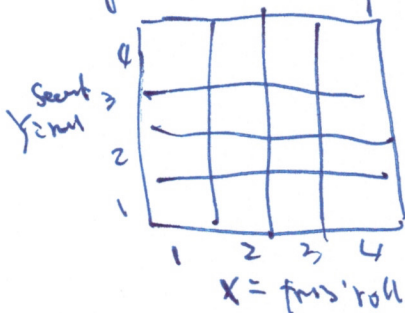
$$\text{then } P(A \cup B) = P(A) + P(B)$$



$$1 = P(A) + P(A^c)$$

$$\Rightarrow P(A) = 1 - P(A^c) \leq 1$$

Probability low: Example



$$- P((X, Y) \text{ is } (1, 1) \text{ or } (1, 2)) = \frac{2}{16}$$

$$- P(\{X=1\}) = \frac{4}{16}$$

$$- P(X+Y \text{ is odd}) = \frac{8}{16}$$

$$- P(\min(X, Y) = 2) = \frac{5}{16}$$

Discrete uniform law.

- let all outcomes be equally likely

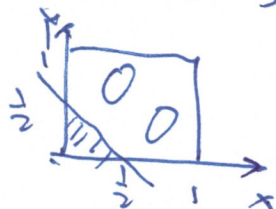
• Then

$$P(A) = \frac{\text{number of elements in } A}{\text{total number of sample points}}$$

- computing probabilities = counting.

Continuous uniform law.

- two "random" numbers in $[0, 1]$



- uniform law:

$$- P(X+Y \leq 1/2) = ? \Rightarrow \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

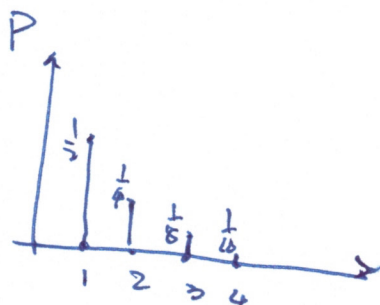
$$- P((X, Y) = (0.5, 0.3)) = 0$$

Probability law: countable infinite sample spaces

- sample space: $\{1, 2, \dots\}$

$$- P(n) = 2^{-n}, n = 1, 2, \dots$$

- find $P(\text{outcome is even})$



$$P(\{2, 4, 6, \dots\}) = P(2) + P(4) + P(6) + \dots = \frac{1}{2^2} + \frac{1}{2^4} + \dots = \frac{1}{3}$$

Countable additivity axiom:

If $A_1, A_2, \dots$ are disjoint events

(sequence of events)

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + \dots$$

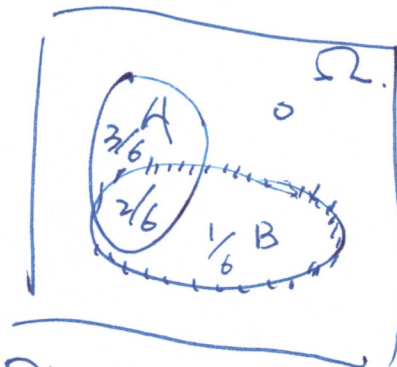
Lecture 2.

Probability doesn't mean it's won't happen, just very rare.
~~in continuous model~~

Conditional probability.

All probabilities are conditional!

- $P(A|B)$ = probability of A
 given that B occurred



- B is our new universe ("sample space")

Definition

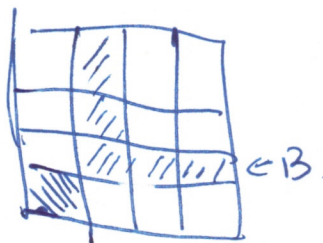
$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{when } P(B) \neq 0$$

$$P(B|B) = 1$$

$$P(A \cap B) = P(B)P(A|B)$$

"B happens occurs, and A occurs after B occurs"

Die roll example



x first roll

$$A \cap B = \emptyset$$

$$P(A \cup B|C) = P(A|C) + P(B|C)$$

1. let B be the event $\min(X, Y) = 2$

• let $M = \max(X, Y)$

$$P(M=1|B) = 0$$

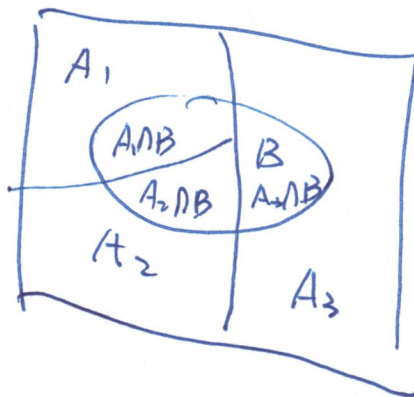
$$P(M=2|B) = \frac{P(M=2 \cap B)}{P(B)} = \frac{1/16}{5/16} = \frac{1}{5}$$

Total probability theorem

- Divide and conquer.
- Partition of sample space into $A_1 A_2 A_3$.
- have $P(B|A_i)$ for every i .

$$P(B) = P(A_1)P(B|A_1) \\ + P(A_2)P(B|A_2) \\ + P(A_3)P(B|A_3)$$

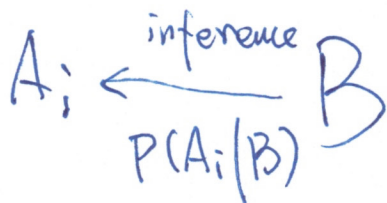
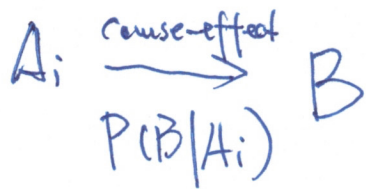
B's probability is the weighted average of each B at different partition.



Bayes rule.

- Prior Probability. $P(A_i)$ - initial "beliefs."
- We know $P(B|A_i)$ for each i .
- Wish to compute $P(A_i|B)$.
 - revise "beliefs", given that B occurred.

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)} \\ = \frac{P(A_i)P(B|A_i)}{P(B)} \\ = \frac{P(A_i)P(B|A_i)}{\sum_j P(A_j)P(B|A_j)}$$



Lecture 3. Independence

Ex. 3 toss of a biased coin

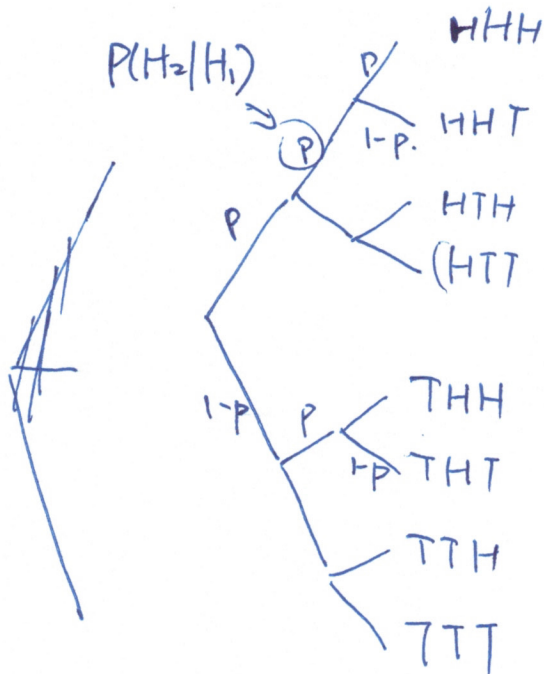
$$P(H) = p \quad P(T) = 1 - p$$

$$P(THT) = P(T) P(H|T) P(T|TH) \\ = (1-p) p (1-p)$$

$$P(1 \text{ head}) = 3 p (1-p)^2$$

$$P(\text{first toss is head} | 1 \text{ head})$$

$$= \frac{P(1 \text{st toss head}, 1 \text{ head})}{P(1 \text{ head})} = \frac{P(HTT)}{P(1 \text{ head})} = \frac{p(1-p)^2}{3 p (1-p)^2} = \frac{1}{3}$$



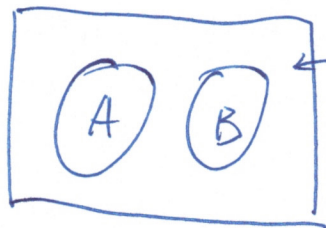
Independent of two events

Definition: $P(B|A) = P(B)$

$$- P(A \cap B) = P(B|A) P(A) \Rightarrow$$

- "occurrence of A provide no info about B's occurrence"

$\Rightarrow P(A \cap B) = P(B) P(A)$ if A and B are independence.
symmetric, and works even if $P(A) = 0$



A and B are not independent!

"A happens $\rightarrow$ B won't happen!"

Conditional probability

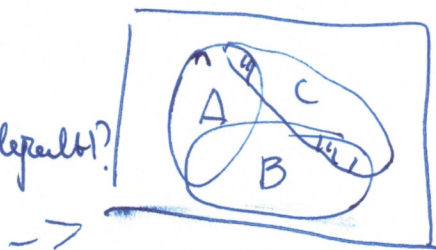
$$P(A \cap B | C) = P(A | C) P(B | C)$$

"in a condition universe C instead of Ω "

Assume A and B are independent.

if C happened, are A and B independent?

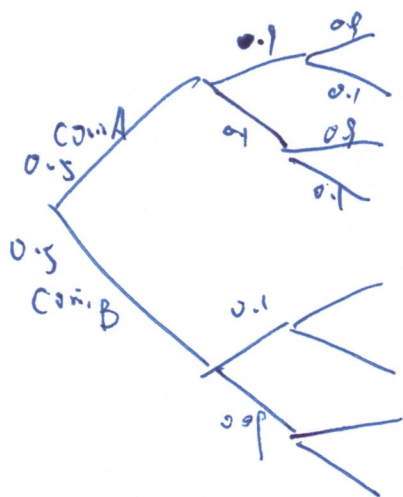
No



$P(A | C)$ and $P(B | C)$ are disjoint.

Conditioning may affect independence.

$P(H | \text{coin A}) = 0.9$, $P(H | \text{coin B}) = 0.1$ two unfair coin A and B
choose either coin with equal probability.



1) when we know it's coin A, are tosses independent? Yes.

2) if we don't know which coin it is, are tosses independent? **No**

- compare:

$$P(\text{toss II} = H) = \frac{1}{2} \cdot 0.9 + \frac{1}{2} \cdot 0.1 = 0.5$$

$$P(\text{toss II} = H | \text{first tosses are head}) \approx 0.9$$

Independent of a collection of events.

- Intuitive definition: information of some of the events tells nothing about probabilities of the remaining events.

Eg.

$$P(A_1 \cap (A_2^c \cup A_3) | A_5 \cap A_6^c) = P(A_1 \cap (A_2^c \cup A_3))$$

- Mathematical definition

Events $A_1, A_2, \dots, A_n$ are called independent if

$$P(A_i \cap A_j \cap \dots \cap A_q) = P(A_i) P(A_j) \dots P(A_q) \quad (\text{any subset})$$

for any distinct indices $i, j, \dots, q$ chosen from $(1, 2, \dots, n)$

Ex. Two independent fair coin tosses

- A : First toss is H

- B : Second toss is H

- $P(A) = P(B) = 1/2$.

HH $1/4$	HT $1/4$
TH $1/4$	TT $1/4$

← C

- C : First and second toss give the same result.

- $P(C) = 1/2$

- $P(C \cap A) = 1/4$

- $P(A \cap B \cap C) = 1/4$

- $P(C | A \cap B) = 1$

pairwise independence does not

imply independence;

$(A, B), (A, C), (B, C)$
are independent, but (A, B, C)
are not!!!

Lecture 4. Counting.

Discrete uniform law.

- Sample point be equally likely

• Then $P(A) = \frac{\text{num of element in } A}{\text{total number of points}} = \frac{|A|}{|\Omega|}$

Basic counting principle.

- r stages

- n_i choices at stages:

$$n_1 = 4$$

$$n_2 = 2$$

$$n_3 = 3.$$



- number of licence plates. 3 letters, 4 digits
 $26^3 \cdot 10^4$

- license plates if repetition is prohibited

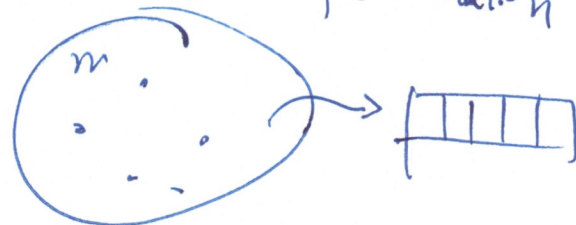
$$26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 \cdot 7.$$

- permutations, number of ways of order elements

$$n!$$

- number of subsets of $(1, \dots, n) =$
 $2^n.$

permutation n



$$\text{ex. } \{1, 2, 3, 4\} \rightarrow (2, 3, 4, 1)$$

$$\{1, 2, 3, \dots, n\}$$

Example.

- probability that roll a ^{fair} die 6 times independently

all given different numbers.

$$\frac{|A|}{|\Omega|} = \frac{6!}{6^6}$$

Combinations

• $\binom{n}{k}$ num of k -element subset from n -elements subset

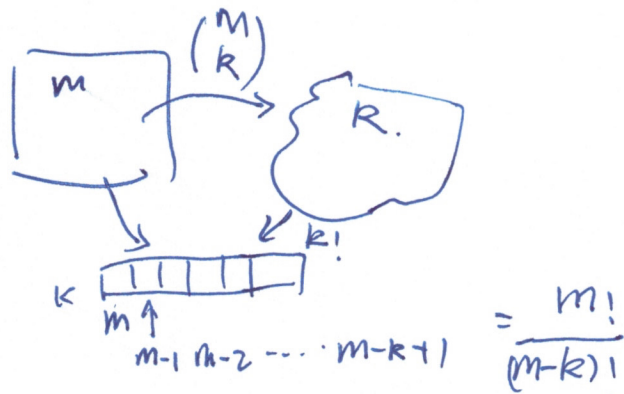
- k distinct items. two ways
 - choose k items one at a time.
 $n(n-1)(n-2)\dots(n-k+1)$
 - Choose k from n , then order them.

$$\binom{n}{k} \cdot k! = \frac{n!}{(n-k)!}$$

$$\binom{n}{k} = \frac{n!}{(n-k)!k!} \leftarrow \text{binomial coefficient.}$$

$$\sum_{k=0}^n \binom{n}{k} = \text{total number of subset} = 2^n.$$

• (all subsets from $0, 1, 2, \dots, n$ elements)



$$\binom{n}{n} = 1 = \frac{n!}{n!0!} = 1$$

$$\binom{n}{0} = 1 = \frac{n!}{0!n!} = 1.$$

• n independent coin toss.

$$P(H) = p$$

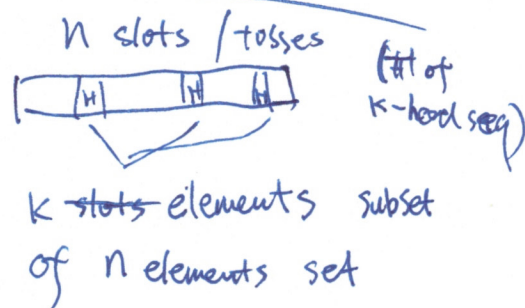
$$P(HTTTHHH) = p(1-p)(1-p)ppp = p^4(1-p)^2.$$

$$P(\text{sequence}) = p^{\# \text{heads}} (1-p)^{\# \text{tails}}.$$

$$P(k \text{ head}) = (\# \text{ of } k\text{-heads seq}) \cdot p^k (1-p)^{n-k}$$

$$= \binom{n}{k} p^k (1-p)^{n-k} \quad k=0, 1, \dots, n.$$

$$\sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = \text{total number of toss} = 1$$



Coin toss problem

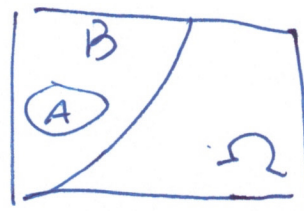
• event B. 3 out of 10 are heads

- Given B occurred, what's the conditional probability the first 2 ^{toss} ~~head~~ are head. (A)

- All outcomes in set B are equally likely. probability $p^3(1-p)^7$

• number of outcomes in B: $\binom{10}{3}$

• out of outcomes in B, ~~what is~~ how many start with HH:



$$\frac{|A \cap B|}{|B|} = \frac{8}{\binom{10}{3}}$$

Partition.

• 52 cards deck, dealt with 4 players

• ~~sample~~ sample space:

$$\frac{52!}{13!13!13!13!}$$

• Event each player get 1 ace.

4.3.2.1

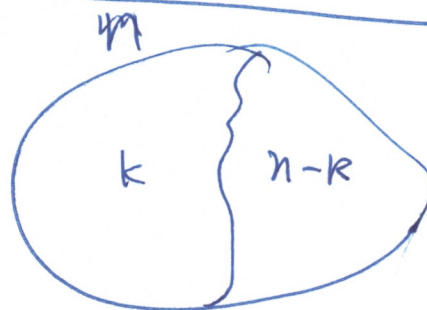
• Remaining cards, each player get 12 cards

$$\frac{48!}{12!12!12!12!}$$

• Answer

$$\frac{4! \cdot 48!}{12!}$$

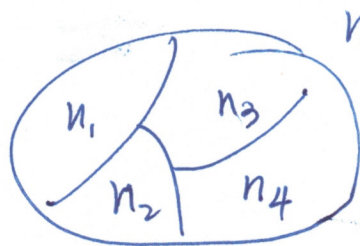
$$\frac{4! \cdot 48!}{12! \cdot 13!13!13!13!}$$



b. binomial

$$\frac{n!}{n_1!n_2!n_3!n_4!}$$

more general



n_1, n_2, n_3, n_4

Lecture 5. Discrete Random Variables.

Random variables

- An assignment of a value (number) to every possible outcome.

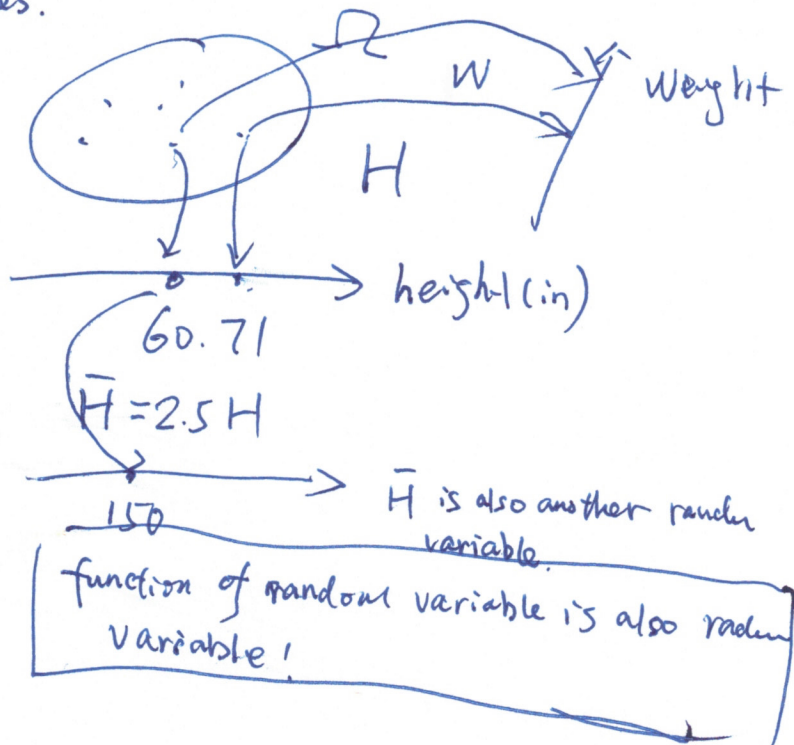
- Mathematically: $\{A \text{ function from the sample space } \Omega \text{ to the real numbers}\}$

- discrete or continuous.

- can have multiple random variables defined on the same sample space.

- Notations =

- random variable $X = \text{a function } \Omega \rightarrow \mathbb{R}$.
- numerical value $x \in \mathbb{R}$.



Probability mass function.

- ("probability distribution" of X)

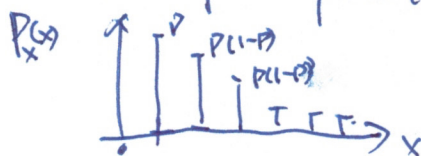
• Notations

$$P_X(x) = P(X=x) = P(\{w \in \Omega \text{ s.t. } X(w)=x\}).$$

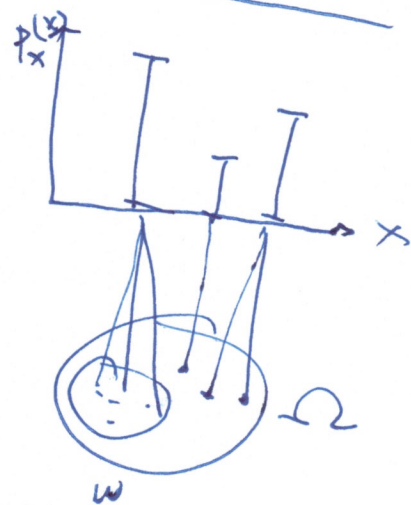
$$P_X(x) \geq 0, \quad \sum_x P_X(x) = 1$$

- Example. X = number of coin tosses until first head.

- assume independent tosses



Geometric Random Variable



$$\begin{aligned} H & X=1 \\ TH & X=2 \\ \underbrace{T \dots T}_k H & X=k. \\ P_X(k) &= (1-p)^{k-1} p \quad k=1, 2, \dots \end{aligned}$$

(X is a function of the outcome)

How to compute $P_X(x)$

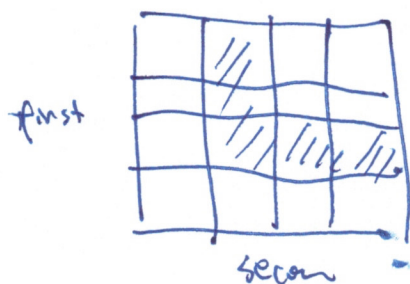
- collect all possible outcomes for which $X=x$
- add their probabilities
- repeat for all x .

Example.: two independent rolls of a fair tetrahedral die.

F : outcome of first roll

S : outcome of second throw.

$X = \min(F, S)$.



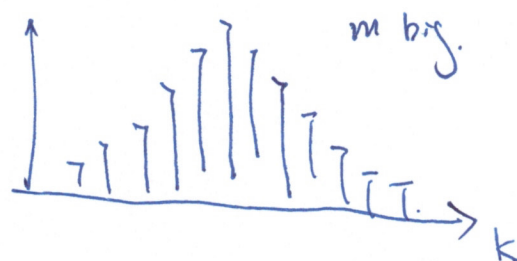
$$P_X(2) = \frac{5}{16}$$

Binomial PMF

- X : number of heads in n independent coin tosses
- $P(H) = p$.
- let $n = 4$.

$$P_X(2) = \binom{4}{2} p^2 (1-p)^2.$$

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}.$$



Expected value of random variable

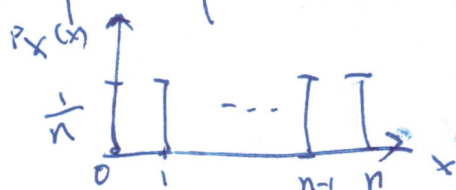
• Definition $E[X] = \sum_x x P_X(x)$

• Interpretation

• Center of gravity of $P_X(x)$.

• Average in large number of repetition of the experiments (to be substitute!)

• Example uniform $0, 1, \dots, n$.



$$E[X] = 0 \cdot \frac{1}{n+1} + 1 \cdot \frac{1}{n+1} + \dots + n \cdot \frac{1}{n+1} = \frac{0+n}{2} \quad (\text{center of gravity due to symmetry})$$

$$\frac{1}{6} \cdot 1 + \frac{1}{2} \cdot 2 + \frac{1}{3} \cdot 4 = 2.5$$

$$\sum_x P_X(x) x$$

Properties of expectation

Let X be a.r.v. and $Y = g(X)$.

- hard: $E[Y] = \sum_y y P_Y(y)$

- Easy: $E[Y] = \sum_x g(x) P_X(x)$ ← need proof

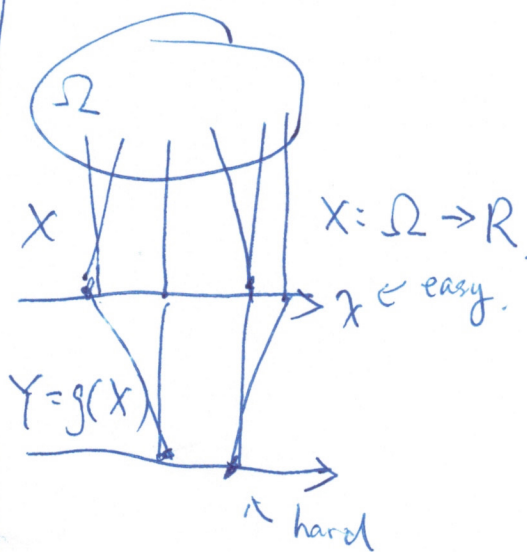
• Caution: In general: $E[g(X)] \neq g(E[X])$
 (Unless g is linear) → (do not commute)

• properties = if α, β are constant,

• $E[a] = a$

• $E[\alpha X] = \alpha E[X]$ [linearity of expectation]

• $E[\alpha X + \beta] = \alpha E[X] + \beta$



$$E[\alpha X] = \sum_x \alpha x P_X(x) = \alpha \sum_x x P_X(x) = \alpha E[X]$$

Variance.

Recall: $E[g(X)] = \sum_x g(x) P_X(x)$

• Second moment = $E[X^2] = \sum_x x^2 P_X(x)$

• Variance: Δ
 $\text{var}(X) = E[(X - E[X])^2]$

$$= \sum_x (x - E[X])^2 P_X(x)$$

$$= E[X^2] - (E[X])^2$$

(average squared distance from the mean!)



Var is measure how far the Value is away from the MEAN value on average!

counter intuitive: a normal distribution a random variable's var is not zero
~~but~~ $\text{var}(X) = 0 \rightarrow P(X=c) = 1$

$$P_X(x) = \begin{cases} 1, & \text{if } x=c \\ 0, & \text{otherwise} \end{cases}$$

Properties:

• $\text{var}(X) \geq 0$

• $\text{var}(\alpha X + \beta) = \alpha^2 \text{var}(X)$

Lecture 6. Discrete Random Variable Examples; Joint PMFs.

- Random variable: Sample space $\rightarrow$ Real value.
- PMF = $p_X(x) = P(X=x)$.

- Expectation: $E[X] = \sum_x x p_X(x)$.
 "center of the mass"
- can't reason on $E \rightarrow E[g(X)] = \sum_x g(x) p_X(x) \neq g(E[X])$ unless g is linear
 $E[\alpha X + \beta] = \alpha E[X] + \beta$

- $E[X - E[X]] = E[X] - E[X] = 0$ "averaged distance from mean is zero"

$\Rightarrow$ var.

$$\text{var}(X) \triangleq E[(X - E[X])^2]$$

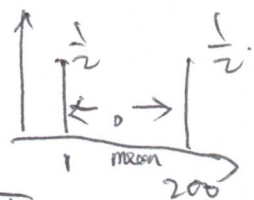
$$= \sum_x (x - E[X])^2 p_X(x)$$

$$= E[X^2] - (E[X])^2$$

var has issue with unit: ie from m $\rightarrow$ m².

Standard deviation: $\sigma_X = \sqrt{\text{var}(X)}$ $\swarrow$ std.

Random speed.
 Travel speed, V



$\alpha = 200$. $T = t(V) = 200/V$

$E[V] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 200 = 100.5$

$\text{var}[V] = \frac{1}{2} (1 - 100.5)^2 + \frac{1}{2} (200 - 100.5)^2 \approx 100^2$

$= \sqrt{100^2} = 100$ (distribution of mean from the)

$E[T] = E[t(V)]$
 $= \frac{1}{2} \cdot 200 + \frac{1}{2} \cdot 1 = 100.5$

$E[TV] = 200 \neq E[T]E[V]$

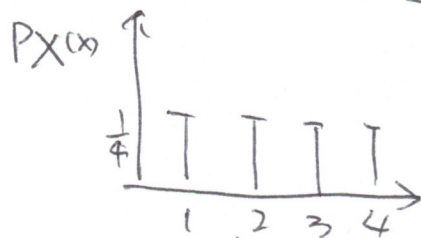
$E[200/V] = E[T]$
 $\neq 200/E[V]$

non linear. can't reason about it!!!

Conditional PMF and expectation

$$P_{X|A}(x) \triangleq P(X=x|A)$$

$$E[X|A] \triangleq \sum_x x P_{X|A}(x)$$



$$\text{Let } A = \{X \geq 2\}$$

$$P_{X|A}(x) = 1/3 \quad x=2,3,4$$

$$E[X|A] = \frac{1}{3}2 + \frac{1}{3}3 + \frac{1}{3}4 = 9$$

(conditional Expectation and PMF are just normal Expectation and PMF)

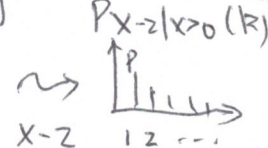
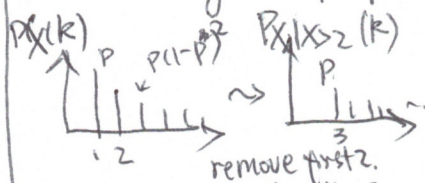
Geometric PMF

X : number of tosses until first head:

$$P_X(k) = (1-p)^{k-1} p, \quad k=1,2,\dots$$

$$E[X] = \sum_{k=1}^{\infty} k P_X(k) = \sum_{k=1}^{\infty} k (1-p)^{k-1} p$$

Memoryless property:



$$\frac{p(1-p)^2}{1-p-p(1-p)} = p$$

$$P(X=2|X>2)(k) = P_X(k)$$

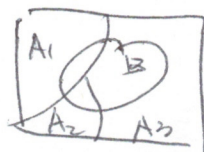
Y : flip until first head.

X : first 2 flip is tail, then count until first head:

Total Expectation theorem

Partition of sample space into disjoint events

$$A_1, A_2, \dots, A_n$$



$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_n)P(B|A_n)$$

$$P_X(x) = P(A_1)P_{X|A_1}(x) + P(A_2)P_{X|A_2}(x) + \dots + P(A_n)P_{X|A_n}(x)$$

PMF is just probability!

(so it applies to conditional PMF)

$$\sum_x x (\text{left}) = \sum_x x (\text{right})$$

$$E[X] = P(A_1)E[X|A_1] + P(A_2)E[X|A_2] + \dots + P(A_n)E[X|A_n]$$

Geometric example.

$$A_1 = \{X=1\}, \quad A_2 = \{X>1\}$$

$$E[X] = P(X=1)E[X|X=1] + P(X>1)E[X|X>1]$$

$$E[X=1] = 1$$

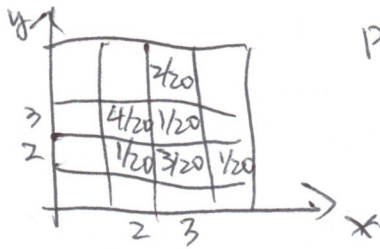
$$P(X=1) = p, \quad P(X>1) = (1-p)$$

$$E[X|X>1] = E[X-1|X-1>0] + 1$$

$$E[X-1|X-1>0] = E[X]$$

Joint PMF

- $P_{X,Y}(x,y) = P(X=x \text{ and } Y=y)$



$$P_{X,Y}(2,3) = 4/20.$$

$$P_X(3) = 6/20$$

- $\sum_x \sum_y P_{X,Y}(x,y) = 1$ $\Omega \rightarrow \mathbb{R}^2$

- $P_X(x) = \sum_y P_{X,Y}(x,y)$ (marginal PMF)

- $P_{X|Y}(x|y) = P(X=x | Y=y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$

(a function of two arguments x, y , but think as the conditional probability of X when y is fixed).

i.e. $P_{X|Y}(x|2) = P(X=x | Y=2) = \begin{cases} 0 & x=1 \\ \frac{1}{5} & x=2 \\ \frac{2}{5} & x=3 \\ \frac{1}{5} & x=4 \end{cases}$

- $\sum_x P_{X|Y}(x|y) = 1.$

Lecture 7. Multiple discrete Random Variables.

Review:

$$P_X(x) = P(X=x) \quad = \text{marginal}$$

$$P_{X,Y}(x,y) = P(X=x, Y=y) \quad = \text{joint}$$

$$P_{X|Y}(x|y) = P(X=x | Y=y) \quad = \text{conditional}$$

$$\sum_x P_{X|Y}(x|y) = 1$$

$$P_X(x) = \sum_y P_{X,Y}(x,y)$$

$$P_{X,Y}(x,y) = P_X(x) P_{Y|X}(y|x)$$

$$\leftarrow P(A \cap B) = P(A) P(B|A)$$

$\uparrow$ $\nwarrow$
 $\{X=x\}$ $\{Y=y\}$

Three random variables.

$$P_{X,Y,Z}(x,y,z)$$

$$P_X(x) = \sum_{y,z} P_{X,Y,Z}(x,y,z)$$

("fix x, find all y and z's and add them up")

$$P_{X,Y,Z}(x,y,z) = P_X(x) P_{Y|X}(y|x) P_{Z|X,Y}(z|x,y)$$

$\uparrow$ same.

$$P(A \cap B \cap C) = P(A) P(B|A) P(C|B,A)$$

• Random variables X, Y, Z are independent:

$$P_{X,Y,Z}(x,y,z) = P_X(x) \cdot P_Y(y) \cdot P_Z(z)$$

for all x, y, z

$$P_{X|Y}(x|y) = P_X(x) \quad \left(\begin{array}{l} P_X(x) = \text{old belief.} \\ y \text{ happened, doesn't change} \\ P_{X|Y}(x|y) \end{array} \right)$$

intuitively.

y \ x	1	2	3	4
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		

1. independent? **NO** = because tells me y gives info about x $\Rightarrow y=1 \rightarrow x=?$
 $y=3 \rightarrow x=?$

2. what if condition **on** $X \leq 2, Y \geq 3$.

Yes

	1	2
4	1/20	2/20
3	2/20	4/20

$\times$

	1	2
3	1/9	2/9
2	2/9	4/9

$$A = \{X \leq 2, Y \geq 3\}$$

$$P_{X,Y|A}(x,y)$$

$$\rightarrow P_{X|Y,A}(x|y) = P_{X|A}(x)$$

Expectations

$$E[X] = \sum_x x P_X(x)$$

$$E[g(X, Y)] = \sum_x \sum_y g(x, y) P_{X, Y}(x, y)$$

In general: $E[g(X, Y)] \neq g(E[X], E[Y])$

Exceptions: (linearity)

$$E[\alpha X + \beta] = \alpha E[X] + \beta$$

$$E[X + Y + Z] = E[X] + E[Y] + E[Z]$$

Independence: If X, Y are independent

$$E[X \cdot Y] = E[X] \cdot E[Y]$$

$$E[g(X)h(Y)] = E[g(X)] \cdot E[h(Y)]$$

(intuitively: X, Y independent $\Rightarrow$ ^(no information) X and $h(Y)$ independent, $\Rightarrow$ $g(X)$ and $h(Y)$ are independent)

independent $\Rightarrow$ no information of, if we know X .

$$\hookrightarrow E[XY] = \sum_x \sum_y x \cdot y P_{X, Y}(x, y)$$

$$= \sum_x \sum_y x \cdot y P_X(x) P_Y(y)$$

$$= \sum_x x P_X(x) \sum_y y P_Y(y)$$

$$= E[X] E[Y]$$

Variance

$$\text{Var}(\alpha X) = \alpha^2 \text{Var}(X)$$

$$\text{Var}(X + \alpha) = \text{Var}(X)$$

$$\text{Let } Z = X + Y$$

only if X, Y are independent.

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Example:

$$\text{If } X = Y \quad \text{Var}(X + Y) = \text{Var}(2X) = 4 \text{Var}(X)$$

$$\text{If } X = -Y \quad \text{Var}(X + Y) = \text{Var}(0) = 0$$

$$\text{If } X, Y \text{ independent. } Z = X - 3Y$$

$$\begin{aligned} \text{Var}(Z) &= \text{Var}(X) + \text{Var}(-3Y) \\ &= \text{Var}(X) + 9 \text{Var}(Y) \end{aligned}$$

Binomial mean and variance.

• $X = \#$ of success in n independent trials

— probability of success p

$$E[X] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} = np.$$

• $X_i = \begin{cases} 1, & \text{if success in trial } i \\ 0, & \text{otherwise.} \end{cases}$ (indicator variables)

$$\sum_i X_i = X$$

• $E[X_i] = 1 \cdot p + 0 \cdot (1-p) = p$

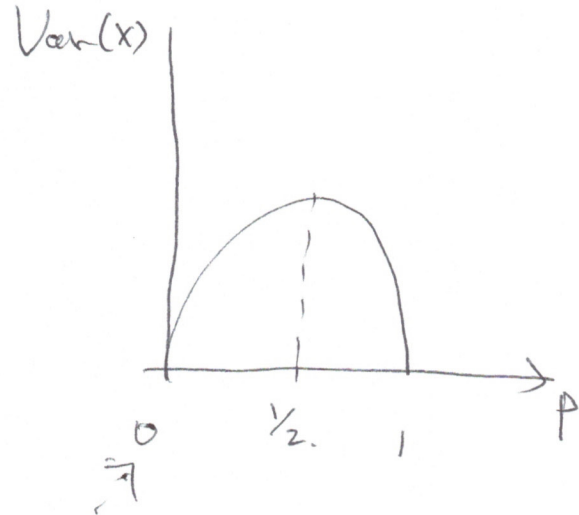
• $E[X] = n \cdot p$

• $Var(X_i) = \overset{\text{prop. 1}}{\underbrace{p(1-p)^2 + (1+p)(0-p)^2}}_{\substack{\uparrow \\ (p-E)^2}}$

$$E[X_i^2] - E[X_i]^2 = E[X_i] - E[X_i]^2 = p(1-p)$$

• $Var(X) = \sum_n Var(X_i) = np(1-p)$

$\uparrow$
 X_i are independent!



The hat problem.

- n people throw their hat in a box and pick one at random.
- X : number of people who get their own hat.

• Find $E[X]$

$$X_i = \begin{cases} 1 & \text{if } i \text{ selects own hat} \\ 0 & \text{otherwise.} \end{cases}$$

- $X = X_1 + X_2 + \dots + X_n$.
- $P(X_i = 1) = 1/n$.
- $E[X_i] = 1 \cdot 1/n + 0 \cdot (1 - 1/n) = 1/n$.
- Are the X_i independent? Yes.

$$\star E[X] = \sum_i E[X_i] = n \cdot \frac{1}{n} = 1.$$

↑
always true even if X_i are dependent

Variance in the hat problem:

$$\bullet \text{Var}(X) = E[X^2] - (E[X])^2 = E[X^2] - 1$$

$$\begin{aligned} X^2 &= \left(\sum_i X_i \right)^2 \\ &= \sum_i X_i^2 + \sum_{i,j: i \neq j} X_i X_j \end{aligned}$$

$$\bullet E[X^2] = E[X_i] = 1/n. \Rightarrow E[X^2] = \sum E[X_i] = n \cdot \frac{1}{n} = 1$$

$$\begin{aligned} \bullet P[X_1, X_2 = 1] &= P(X_1 = 1) P(X_2 = 1 | X_1 = 1) \\ &= \frac{1}{n} \cdot \frac{1}{n-1} = E[X_1 X_2] \end{aligned}$$

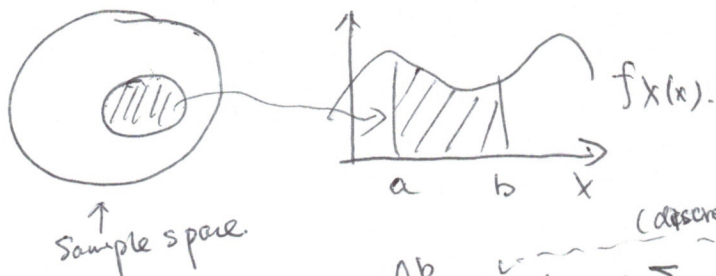
$$\bullet E[X^2] = \sum_i E[X_i^2] + \sum \sum E[X_i X_j] = n \cdot \frac{1}{n} + (n^2 - n) \frac{1}{n} \frac{1}{n-1}$$

$$\sim \text{Var}(X) = E[X^2] - 1 = 2 - 1 = 1$$

Lecture 8. Continuous random variable.

Continuous r.v.'s and pdf's

A continuous r.v. is described by a probability density function f_X



$$P(a \leq X \leq b) = \int_a^b f_X(x) dx \quad \text{(discrete)} \quad \sum_{x \in [a,b]} P_X(x)$$

$$\textcircled{1} \int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$\textcircled{2} f_X(x) > 0$$

$$\textcircled{3} f_X(x) \text{ can } > 1 \text{ since it's not probability.}$$

$$P(x \leq X \leq x+\delta) = \int_x^{x+\delta} f_X(s) ds \approx f_X(x) \cdot \delta \quad \text{small interval}$$

"density is probability per unit length with small δ ".

$$P(X \in B) = \int_B f_X(s) ds.$$

continuous r.v. still maps from $\Omega \rightarrow \mathbb{R}$, where

$P(a \leq X \leq b)$ are the probability of samples

$\{\omega: a \leq X(\omega) \leq b\}$ in the sample space Ω .

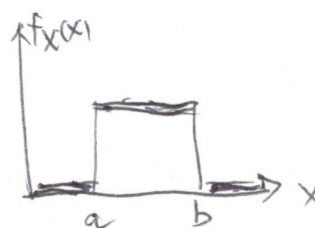
Means and Variance.

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$$\begin{aligned} \text{var}(X) &= \int_{-\infty}^{\infty} (x - E[X])^2 f_X(x) dx \\ &= E[X^2] - (E[X])^2 \end{aligned}$$

Continuous Uniform r.v.



$$f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b, \quad 0, \text{ otherwise}$$

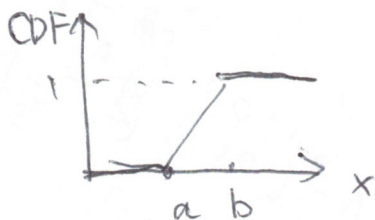
$$E[X] = \int_a^b x \frac{1}{b-a} dx = \frac{b+a}{2}$$

$$E[X^2] = \int_a^b \left(x - \frac{a+b}{2}\right)^2 \frac{1}{b-a} dx = \frac{(b-a)^2}{12}$$

$$\sigma_X = \frac{b-a}{\sqrt{12}}$$

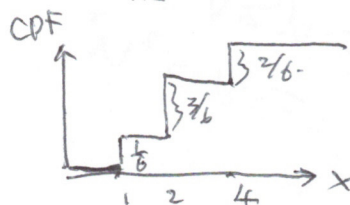
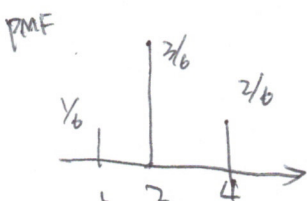
Cumulative distribution function (CDF)
unify discrete and continuous r.v.

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt \quad \left| \quad \frac{dF_X}{dx}(x) = f_X(x) \right|$$



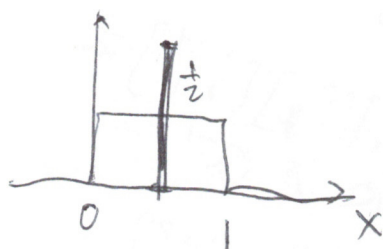
• Also for discrete r.v.s.

$$F_X(x) = P(X \leq x) = \sum_{k \leq x} P_X(k)$$



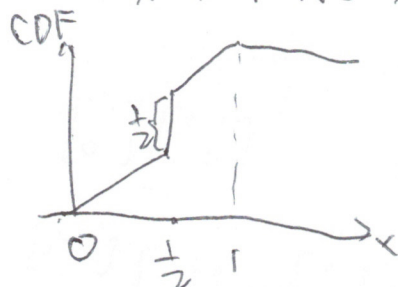
• Mixed distribution:

CDF can work with mixed random variable.



• the corresponding CDF:

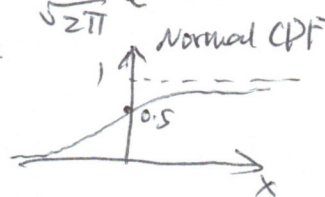
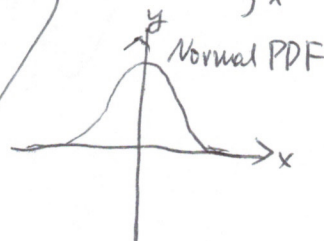
$$F_X(x) = P(X \leq x)$$



Gaussian (normal) distribution.

• Standard normal $N(0,1)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$



$$E[X] = 0, \quad \text{var}(X) = 1$$

• General normal $N(\mu, \sigma^2)$

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

• It turns out that:

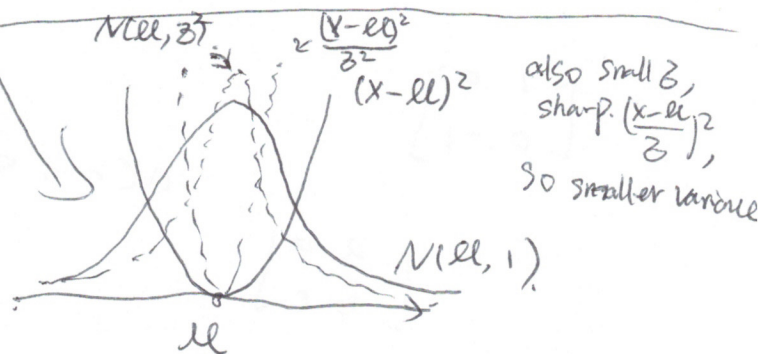
$$E[X] = \mu, \quad \text{var}(X) = \sigma^2$$

• Let $Y = aX + b$

$$E[Y] = a\mu + b, \quad \text{var}(Y) = a^2\sigma^2$$

★ Fact: $Y \sim N(a\mu + b, a^2\sigma^2)$ [linearly of N]

"intuitively: a change the scale of the bell shape, b move it on x axes"



calculate normal probability.

No closed form for CDF.

- but there are tables of $N(0,1)$.

If $X \sim N(\mu, \sigma^2)$ then $\frac{X - \mu}{\sigma} \sim N(0,1)$.

If $X \sim N(2, 16)$

$$P(X \leq 3) = P\left(\frac{X-2}{4} \leq \frac{3-2}{4}\right) = \text{CDF}(0.25)$$

↑
why it works?

★ The constellation of concepts

discrete

continuous

$$P_X(x)$$

$$F_X(x)$$

$$f_X(x)$$

$$P_{X,Y}(x,y)$$

$$E[X], \text{var}(X)$$

$$f_{X|Y}(x,y)$$

$$P_{X|Y}(x|y)$$

$$f_{X,Y}(x,y)$$

$$\Sigma$$

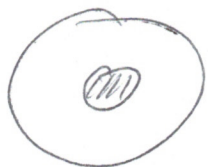
$$\int$$

$$P_X$$

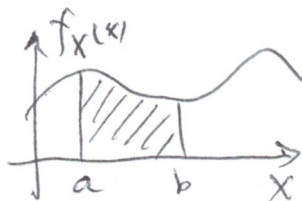
$$f_X$$

Lecture 9. Multiple Continuous Random variable.

Continuous r.v. and pdf's



event $[a \leq x \leq b]$



$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

• intuitively: little interval.

$$P(x \leq X \leq x+\delta) \approx f_X(x) \cdot \delta$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$\downarrow$
 δx

$$\begin{aligned} f_X &\geq 0 \\ \int_{-\infty}^{\infty} f_X(x) dx &= 1 \\ P(X=a) &= 0 \end{aligned}$$



event $\{w: M(w) \in S\}$

Joint PDF. $f_{X,Y}(x,y)$.

$$P((X,Y) \in S) \triangleq \iint_S f_{X,Y}(x,y) dx dy$$

• Interpretation (probability of unit "2D" area)

$$P(x \leq X \leq x+\delta, y \leq Y \leq y+\delta) \approx f_{X,Y}(x,y) \delta^2$$

• Expectation.

$$E[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{X,Y}(x,y) dx dy$$

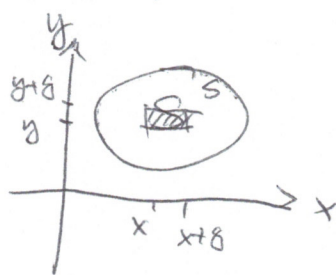
• From joint to marginal.

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy \quad ?$$

• X and Y are called independent if

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \text{ for all } x,y$$

$$\Rightarrow P(X \in A, Y \in B) = P(X \in A) P(Y \in B)$$



$f_{X,Y}$ is pdf of $M: \Omega \rightarrow \mathbb{R}^2$

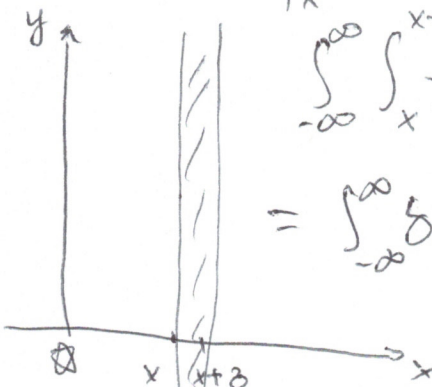
$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y} = 1$$

$$f_{X,Y} \geq 0$$

$$f_X(x) \cdot \delta = P(x \leq X \leq x+\delta) =$$

$$\int_{-\infty}^{\infty} \int_x^{x+\delta} f_{X,Y}(x,y) dx dy$$

$$= \int_{-\infty}^{\infty} \delta f(x,y) dy$$



intuition:

$f_{X,Y}(x,y) \rightarrow$ height (x,y)

$f_X(x) \rightarrow$ area at x

$P(X \in A, Y \in B) \rightarrow$ volume

Buffon's needle.

• Parallel lines at distance d .

Needle of length l ($l < d$).

• Find P (needle intersects one of the lines)

• $X \in [0, d/2]$ = distance of needle midpoint to nearest line

• Model: X, θ uniform, independent.

$$f_{X, \theta}(x, \theta) = (2/d)(2/\pi) \quad 0 \leq x \leq d/2, \quad 0 \leq \theta \leq \pi/2$$

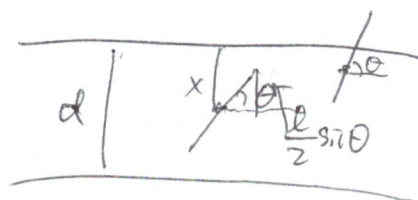
$$f_X(x) f_\theta(\theta) = (2/d) \cdot (2/\pi) \quad \text{"because independent, and uniform"}$$

• Intersect if $X \leq \frac{l}{2} \sin \theta$

$$P(X \leq \frac{l}{2} \sin \theta) = \int \int_{x \leq \frac{l}{2} \sin \theta} f_X(x) f_\theta(\theta) dx d\theta$$

$$= \frac{4}{\pi d} \int_0^{\pi/2} \int_0^{\frac{l}{2} \sin \theta} dx d\theta$$

$$= \frac{4}{\pi d} \int_0^{\pi/2} \frac{l}{2} \sin \theta d\theta = \frac{2l}{\pi d}$$



Conditioning.

Recall

$$P(x \leq X \leq x+\delta) \approx f_X(x) \cdot \delta$$

By analogy

$$P(x \leq X \leq x+\delta | Y \approx y) \approx f_{X|Y}(x|y) \cdot \delta$$

approximate! because $P(Y=y)=0$, so we say Y is "infinitesimally close" to y (or take limit).

This leads us to the definition

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \quad \text{if } f_Y(y) > 0$$

For given y , conditional PDF is a (normalized) "section" of the joint PDF.

If independent, $f_{X,Y} = f_X f_Y$.

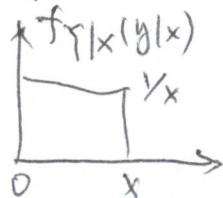
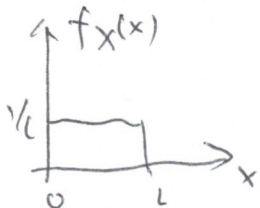
$$f_{X|Y}(x|y) = f_X(x).$$

Stick breaking example.

Break a stick of length l twice.

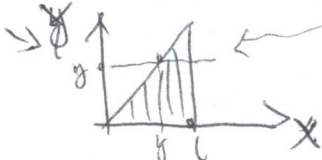
1) break at X : uniform in $[0, l]$

2) break at Y : uniform in $[0, X]$



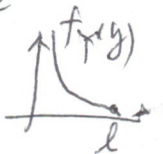
$$\begin{aligned} f_{X,Y}(x,y) &= f_X(x) f_{Y|X}(y|x) \\ &= \frac{1}{l} \cdot \frac{1}{x} \quad 0 \leq Y \leq X \leq l \end{aligned}$$

on the set



$$\begin{aligned} E[Y|X=x] &= \int y f_{Y|X}(y|x) dy \\ &= \int_0^x y \cdot \frac{1}{x} dy = \frac{x}{2} \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \int f_{X,Y}(x,y) dx \\ &= \int_y^l \frac{1}{lx} dx = \frac{1}{l} \log \frac{l}{y} \quad 0 \leq y \leq l \end{aligned}$$



$$\begin{aligned} E[Y] &= \int_0^l y f_Y(y) dy = \int_0^l y \frac{1}{l} \log \frac{l}{y} dy \\ &= \frac{l}{4} \end{aligned}$$

Lecture 10. Continuous Bayes Rule.

(inference problem)

The Bayes Variation

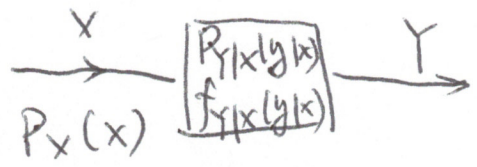
$$P_{X|Y}(x|y) = \frac{P_{X,Y}(x,y)}{P_Y(y)} = \frac{P_X(x)P_{Y|X}(y|x)}{P_Y(y)}$$

— Continuous counterpart

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{f_X(x)f_{Y|X}(y|x)}{f_Y(y)}$$

$$f_Y(y) = \int_x f_X(x)f_{Y|X}(y|x)dx$$

"initial belief $f_X(x)$, we measure $f_{Y|X}(y|x)$, then update belief of X to $f_{X|Y}(x|y)$."



$$P_X(x)$$

$$P_{X|Y}(x|y)$$

$$f_{X|Y}(x|y)$$

$$P_X(x)P_{Y|X}(y|x) = P_{X,Y}(x,y)$$

$$= P_Y(y)P_{X|Y}(x|y)$$

— Discrete X , continuous Y

$$P_{X|Y}(x|y) = \frac{P_X(x)f_{Y|X}(y|x)}{f_Y(y)}$$

$$f_Y(y) = \sum_x P_X(x)f_{Y|X}(y|x)$$

Example:

• X : a discrete signal, "prior" $P_X(x)$

• Y : noisy version of X

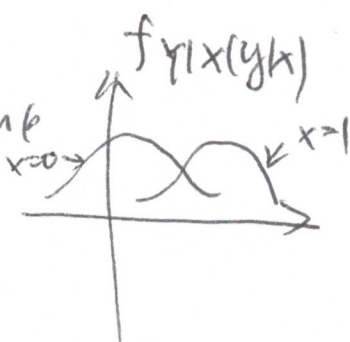
$f_{Y|X}(y|x)$: continuous noise model.

$$X = 0, 1$$

$$f_{Y|X}(y|x) \sim \text{Normal}$$

$$Y = X + W$$

Gaussian noise



$$P(X=x, y \leq Y \leq y+\delta)$$

$$= P(X=x)P(y \leq Y \leq y+\delta | X=x) \textcircled{1}$$

$$= P(y \leq Y \leq y+\delta)P(X=x | y \leq Y \leq y+\delta) \textcircled{2}$$

$$\textcircled{2} = \textcircled{1} \Rightarrow P_X(x)f_{Y|X}(y|x) \cdot \delta$$

$$= f_Y(y)P_{X|Y}(x|y)$$

- Continuous X , discrete Y

$$f_{X|Y} = \frac{f_X(x) P_{Y|X}(y|x)}{P_Y(y)}$$

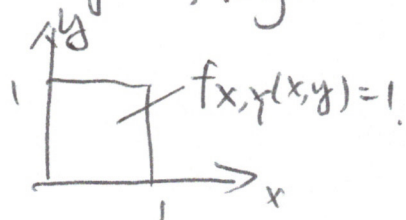
$$P_Y(y) = \int_x f_X(x) P_{Y|X}(y|x) dx.$$

Example:

- X : a continuous signal, "prior" $f_X(x)$.
(e.g. intensity of light beam)
- Y : discrete r.v. affected by X .
(e.g. photon count)
- $P_{Y|X}(y|x)$: model of the discrete r.v.

What is derived distribution.

- It is a PMF or PDF of a function of one or more random variables with known probability law, E.g.



- Obtaining the PDF for $g(X,Y) = Y/X$

involves deriving a distribution.

Note: $g(X,Y)$ is a random variable.

When to not find them?

- Don't need PDF if only want to compute expected value:

$$E[g(X,Y)] = \iint g(x,y) f_{X,Y}(x,y) dx dy$$

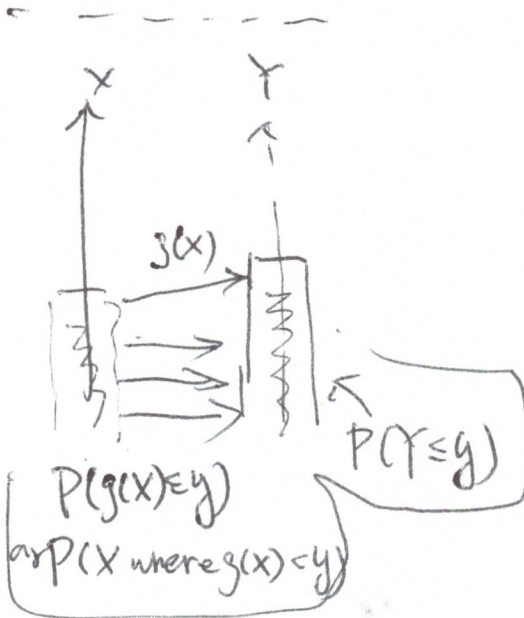
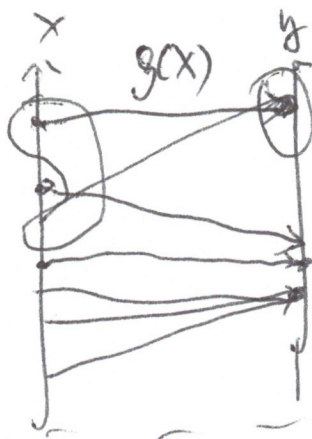
How to find them.

• Discrete case:

- Obtain probability mass for each possible values of $Y=g(X)$

$$P_Y(y) = P(g(X)=y)$$

$$= \sum_{x:g(x)=y} P_X(x)$$



• For continuous case=

• Two-step procedure:

- Get CDF of $Y: F_Y(y) = P(Y \leq y)$

- Differentiate to get $\frac{dF_Y}{dy}(y)$

Example:

• X : uniform on $[0, 2]$

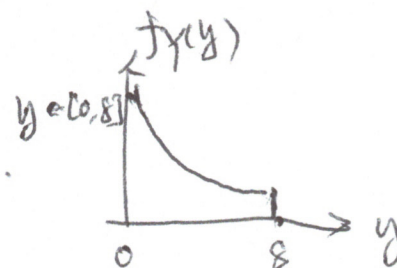
• Find PDF of $Y=X^3$ $Y \in [0, 8]$

• Solution

$$F_Y(y) = P(Y \leq y) = P(X^3 \leq y)$$

$$= P(X \leq y^{1/3}) = \frac{1}{2} y^{1/3}$$

$$f_Y(y) = \frac{dF_Y}{dy}(y) = \frac{1}{6} y^{-2/3} \quad y \in [0, 8]$$



Example

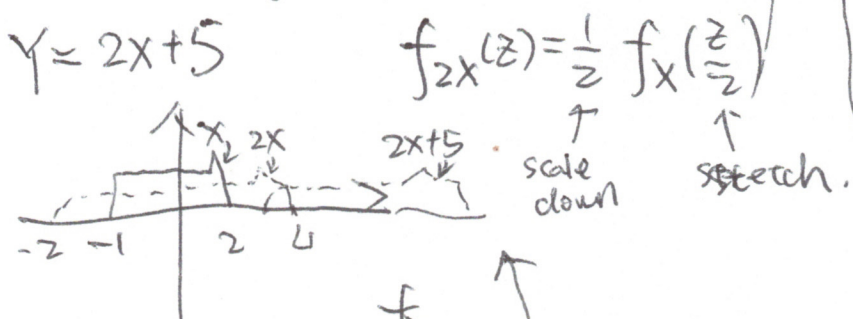
- speed uniform between 30-60
- distance 200
- distribution of the duration T .

$$T = \frac{200}{V}$$

The pdf of $Y = aX + b$

$$(a=2 \text{ } b=5)$$

$$Y = 2X + 5$$



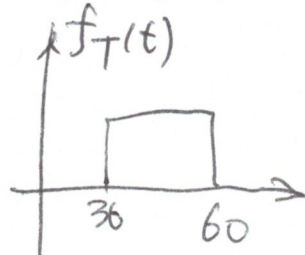
$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

(a) stretch
(b) move

$$F_T(t) = P(T \leq t) = P\left(\frac{200}{V} \leq t\right) = P(V \geq \frac{200}{t})$$

$$= \frac{1}{30} \left(60 - \frac{200}{t}\right)$$

$$f_T(t) = \frac{200}{30} \frac{1}{t^2}, \frac{200}{60} \leq t \leq \frac{200}{30}$$



Proof.

$$a > 0. \quad F_Y(y) = P(Y \leq y)$$

$$= P(aX + b \leq y)$$

$$= P(X \leq \frac{y-b}{a})$$

$$= F_X\left(\frac{y-b}{a}\right)$$

~~$$f_Y(y) = f_X\left(\frac{y-b}{a}\right)$$~~

~~$$\frac{dF_Y}{dy} = \frac{dF_X\left(\frac{y-b}{a}\right)}{d\left(\frac{y-b}{a}\right)} \cdot \frac{y-b}{a}$$~~

$$\frac{dF_Y(y)}{dy} = \frac{dF_X\left(\frac{y-b}{a}\right)}{d\left(\frac{y-b}{a}\right)} \cdot \frac{y-b}{a}$$

$$= \frac{dF_X(u)}{du} \frac{du}{dx} = f_X(u) \frac{1}{a} = \frac{1}{a} f_X\left(\frac{y-b}{a}\right)$$